

Customer Agent

Role: Notifier Service · Pipeline Position: Step 5 of 5 · Endpoint: POST
/customer/notify

Generates ETA notifications, logs alerts, and sends HTML confirmation emails to customers.

Overview

The Customer Agent is the fifth and final agent in the pipeline. It loads trip attributes, formats an ETA string, composes a customer dispatch message, logs it in the notifications table, and sends a rich HTML email with a Google Maps tracking link.

Responsibilities

- Fetch trip details (source, destination, estimated_duration).
- Retrieve assigned driver name and phone number.
- Retrieve assigned vehicle plate number.
- Format ETA string from estimated_duration minutes.
- Compose customer dispatch message.
- Insert notification log into the notifications table (type: Dispatch_Notice).
- Send HTML email to customer with Google Maps tracking link.

Email Content

Email Field	Content
Origin	Source city / location
Destination	Destination city / location
Route Distance	Distance in km
ETA	Estimated arrival time (e.g., 7h 42m)
Driver Name	Assigned driver full name
Driver Phone	Contact phone number
Vehicle Plate	Vehicle registration number

Tracking Button	Google Maps directions link
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API Endpoint

Method	Path	Description
POST	/customer/notify	Generate and send customer delivery notification

Request Payload

```
{ "trip_id": "<uuid>" }
```

Response Payload

```
{
  "status": "success",
  "data": {
    "trip_id": "<uuid>",
    "customer_message": "Your shipment has been dispatched. Driver Speedy...",
    "notification_type": "Trip Update"
  }
}
```

Internal Logic Flow

```
notify_customer(trip_id)
  ■
  ■■■ 1. get_trip(trip_id)           → AgentFleetException if not found
  ■■■ 2. get_driver(driver_id)      → name, phone
  ■■■ 3. get_vehicle(vehicle_id)    → vehicle_number
  ■■■ 4. Format eta_str from estimated_duration minutes
  ■■■ 5. Compose customer_message string
  ■■■ 6. insert_notification(Dispatch_Notice)
  ■■■ 7. send_email_async(customer_email, html_body)
```

Module Files

File	Purpose
agent.py	Agent registry entry — wraps service call

service.py	Business logic: message construction, notification log, email
repository.py	DB queries: trip, driver, vehicle fetch, notification insert
routes.py	FastAPI router — POST /customer/notify
schemas.py	Pydantic request/response models
prompts.py	AI prompt templates
tools.py	LLM tool definitions

Error Cases

Condition	Exception	HTTP Code
Trip not found by trip_id	AgentFleetException	400
Notification insert failure	Exception re-raised	500
Email delivery failure	Warning logged only	—